INFOTACT SOLUTIONS · INTERNSHIP PROJECT

CareFlow

Clinical Pathway Process Mining — Complete Build Guide

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Role: Data Associate L1 Intern | **Domain:** Data Analytics
Project window: 10/10/2026 → 10/11/2026 (1 month)
Reference date: 10 October 2026

CareFlow kya hai? (1 minute me samjho)

Hospital me patient ka safar kai steps se guzarta hai — **Registration → Triage → Lab Test → Diagnosis → Treatment → Discharge**. Har patient ka yeh rasta thoda different hota hai: kuch patients ka test dobara hota hai, kuch ko ICU jana padta hai, kuch bahut zyada wait karte hain.

CareFlow ek hospital ke **event log** ko padhta hai (1 row = 1 patient ka 1 step) aur **process mining** lagakar ye nikalta hai:

1. **Discovery** — asli pathway ka process model (Directly-Follows Graph) banata hai
2. **Analytics** — bottleneck, waiting time, cost aur resource load nikalti hai
3. **Conformance** — real cases expected pathway se kitne hatt gaye, woh check karti hai
4. **Prediction** — kaunsa patient long stay (> 24 hours) hoga, woh predict karti hai
5. **Dashboard** — sab kuch interactive Streamlit dashboard me dikhati hai

> **Aapka assignment letter kehta hai:** submit karna hai **Source Code · Project Documentation · Project Report · Presentation (PPT) · Model Evaluation Results · Final Demonstration**. Yeh guide wahi 6 cheezein aapke liye bana deti hai — sab ready hai, bas run karke apne naam se submit kar dijiye.

Yeh guide kaise use karein

#	PDF file	Kya sikhaata hai	Time
Step 1	`CareFlow_Step1_Setup`	Python + VS Code + folder setup	10 min

Step 2	`CareFlow_Step2_Install_Dependencies`	pip install, venv, pm4py, Graphviz	10 min
Step 3	`CareFlow_Step3_Import_Your_Data`	Apna data kaise import karein	10 min
Step 4	`CareFlow_Step4_Run_Pipeline`	Poora pipeline step-by-step chalana	5 min
Step 5	`CareFlow_Step5_Dashboard_And_Demo`	Dashboard + demonstration	10 min
Extra	`CareFlow_Project_Documentation`	Deliverable 2	—
Extra	`CareFlow_Project_Report`	Deliverable 3	—
Extra	`CareFlow_Model_Evaluation_Results`	Deliverable 5	—
Extra	`CareFlow_Presentation_Content`	Deliverable 4 (PPT)	—

0. Sabse pehle: zip file kholna

1. Download ki hui `CareFlow_Project.zip` ko apne computer par **right-click → Extract All** (Windows) ya double-click (Mac) se kholiye.
2. Ek folder banega `CareFlow` — uske andar `src`, `data`, `dashboard`, `outputs`, `docs` sab honge.
3. Is folder ko aisi jagah rakhiye jahan path me space na ho, jaise `C:\Projects\CareFlow` ya `~/Projects/CareFlow`.

> **Important:** Zip ko `Downloads` me hi mat chhodiye aur Desktop par space wale naam se mat rakhiye. Simple path use kijiye — baad me errors kam aayenge.

1. Poora flow ek nazar me

```
  ZIP kholo
     |
     v
[Step 1] Python + VS Code install  ->  CareFlow folder VS Code me open
     |
     v
[Step 2] python -m venv .venv  ->  pip install -r requirements.txt
     |
     v
[Step 3] data/raw/event_log.csv me apna data daalo (ya synthetic data use karo)
     |
     v
[Step 4] python run_all.py       <- 6 steps apne aap chal jaate hain
     |
     v
[Step 5] streamlit run dashboard/app.py   ->  browser me dashboard
     |
     v
  Submit: Source Code + Documentation + Report + PPT + Model Evaluation + Demo
```

2. Kya-kya chahiye (requirements)

Cheez	Minimum	Recommended	Kahan se
Computer	4 GB RAM	8 GB RAM	—
Disk space	1.5 GB free	3 GB free	—
Python	3.10	3.12 / 3.13	`python.org/downloads`
Editor	VS Code	VS Code + Python extension	`code.visualstudio.com`
Internet	pip install ke liye	—	—

Gyaan: Sirf core pipeline ke liye koi paid software nahi chahiye — sab open-source hai. Optional `pm4py` bhi free hai.

3. Project folder ka structure (yaad rakhiye)

```
CareFlow/
|-- config/config.yaml           <- saari settings yahan (cases, threshold, paths)
|-- data/
|   |-- raw/                     <- apna ya synthetic data
|   `-- processed/               <- cleaning ke baad ka clean data
|-- src/
|   |-- utils.py                 <- config loader + logger
|   |-- generate_data.py         <- STEP 1 event log banata hai
|   |-- preprocess.py            <- STEP 2 cleaning + features
|   |-- discover_models.py       <- STEP 3 process discovery
|   |-- analyze_insights.py      <- STEP 4 KPI + bottleneck + insights
|   |-- conformance.py           <- STEP 5 fitness checking
|   `-- ml_model.py              <- STEP 6 long-stay prediction
|-- dashboard/app.py             <- STEP 7 Streamlit dashboard
|-- outputs/
|   |-- figures/                 <- 14 charts (PNG)
|   |-- reports/                 <- JSON + CSV results
|   `-- models/                  <- trained model file
|-- docs/                        <- yeh saare guide PDFs
|-- run_all.py                   <- one-click runner
|-- requirements.txt
`-- README.md
```

4. Reference run ke results (aapko yahi milenge)

Patient cases	1,500
Events (cleaning ke baad)	13,583
Activities	15
Directly-follows edges	28

Pathway variants	14 (top variant = 29.87 %)
Average length of stay	9.65 hours (median 6.14 h)
Long-stay cases (> 24 h)	17 (1.13 %)
Biggest bottleneck	Observation — 226.6 min average wait
Costliest department	Operation Theatre
Average cost per case	INR 17,258
Peak hospital load	Friday, 14:00
Pipeline runtime	~17 seconds

Honest note (viva me zaroor bataiye): reference run **synthetic** data par hua hai — `src/generate_data.py` realistic hospital event log generate karta hai, taaki aap patient-confidential data ke bina poora pipeline demonstrate kar sakein. Isi wajah se synthetic data par conformance fitness **1.0** aata hai. **Asli hospital log** par fitness generally **0.70 – 0.95** ke range me aata hai. Asli data daalne ke liye Step 3 dekhiye.

5. Delieverable checklist

Assignment deliverable	Aapke paas kya hai	Status
Source Code	`src/` + `dashboard/` + `run_all.py` + `config/`	Ready
Project Documentation	`CareFlow_Project_Documentation.pdf`	Ready
Project Report	`CareFlow_Project_Report.pdf`	Ready
Presentation (PPT)	`CareFlow_Presentation_Content.pdf` (13 slides, copy-paste ready)	Ready
Model Evaluation Results	`CareFlow_Model_Evaluation_Results.pdf`	Ready
Final Project Demonstration	`CareFlow_Step5_Dashboard_And_Demo.pdf`	Ready

6. Mentor ko progress update bhejne ka format

```
Subject: CareFlow - Project Progress Update (Week 1 of 4)

Respected Mentor,

I have completed the following on the CareFlow (Clinical Pathway
Process Mining) project:

1. Environment setup and dependency installation - done
2. Event log ingestion, cleaning and validation - done
   (13,664 raw rows -> 13,583 clean events, 81 duplicates removed)
3. Process discovery - DFG with 28 edges and 14 pathway variants
```

```
4. Analytics - LOS 9.65 h average, bottleneck = Observation (226.6 min)
5. Conformance checking and long-stay prediction model - completed
6. Interactive dashboard (Streamlit) - ready for demonstration

Artefacts: source code, project report, documentation and model
evaluation results attached.

Requesting a review call at your convenience.

Regards,
Yash Rahangdale
Data Associate L1 Intern | Ref. No. ITS/DA/1393
```

Ab Step 1 PDF kholiye aur setup shuru kijiye. Har step me exact commands diye hain — copy-paste kar dijiye, kuch yaad rakhne ki zarurat nahi.